

COMP2300 Set 3 Practice Exam Solutions

Part A: Multiple Choice

- A1. C
- A2. D
- A3. B
- A4. B
- A5. B
- A6. C
- A7. B
- A8. D
- A9. B
- A10. B

Part B: Computational and Design Problems

B1. SR Latch

Using NOR SR latch rules:

- (0,0): hold $Q = 0$
- (1,0): set $Q = 1$
- (0,0): hold $Q = 1$
- (0,1): reset $Q = 0$
- (0,0): hold $Q = 0$
- (1,1): invalid (both outputs forced low)
- (0,0): indeterminate due to prior invalid state

B2. D Latch

Transparent when CLK=1; hold when CLK=0.

Interval	1	2	3	4	5
CLK	0	1	1	0	1
D	1	0	1	1	0
Q	0	0	1	1	0

B3. D Flip-Flop

Samples on rising edges. Starting $Q = 1$:

$$Q(t_1) = 0, \quad Q(t_2) = 1, \quad Q(t_3) = 1, \quad Q(t_4) = 0$$

B4. Registers and Memory

- (a) $\log_2(64K) = 16$ address bits.
- (b) Capacity = $64K \times 16$ bits = 1,048,576 bits = 131,072 bytes (128 KiB).
- (c) Byte-addressable capacity = $64K \times 8$ bits = 524,288 bits = 65,536 bytes (64 KiB).

B5. Timing constraints

- (a) $T_c = t_{pcq} + t_{pd} + t_{setup} = 0.4 + 6.0 + 0.2 = 6.6$ ns.
- (b) $f_{max} = 1/T_c \approx 1/6.6$ ns = 151.5 MHz.
- (c) Aperture time = $t_{setup} + t_{hold} = 0.2 + 0.1 = 0.3$ ns.

B6. Pipeline throughput

- (a) $T_c = 0.3 + \max(4, 3, 5) + 0.2 = 5.5$ ns.
- (b) Cycles = $K + (N - 1) = 3 + 19 = 22$. Time = 22×5.5 ns = 121 ns.

Part C: Past Exam Questions**C1. Moore FSM concept**

A Moore FSM is a finite state machine whose outputs depend only on the current state, not directly on inputs. Example: a traffic light controller where each state encodes the light color.

C2. FSM design (three consecutive 1s)

- (a) States represent count of consecutive 1s: S_0 (0), S_1 (1), S_2 (2), S_3 (3 or more). Output $X = 1$ only in S_3 . Transitions: on $A = 1$, $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_3$; on $A = 0$, any state $\rightarrow S_0$.
- (b) Example encoding: $S_0 = 00$, $S_1 = 01$, $S_2 = 10$, $S_3 = 11$.
- (c) Next-state equations (one valid set):

$$D_1 = A \cdot (Q_1 + Q_0), \quad D_0 = A \cdot (Q_1 + \overline{Q_0})$$

Output: $X = Q_1 Q_0$.

- (d) Schematic: input A into next-state logic; state register is two D flip-flops; outputs from register feed output logic for X .

C3. Pipeline timing

- (a) $T_c = 0.2 + \max(3.4, 3.3, 3.7) + 0.1 = 4.0$ ns.
- (b) $K = 3$, $N = 12$ cycles $= 3 + 11 = 14$. Time $= 14 \times 4.0$ ns $= 56$ ns.
- (c) $N = 10^9$ cycles $= 10^9 + 2$. Time ≈ 4.0 ns $\times (10^9 + 2) \approx 4.000000008$ s.
- (d) Four pipelines: 2.5×10^8 tokens each. Time ≈ 4.0 ns $\times (2.5 \times 10^8 + 2) \approx 1.000000008$ s.